Baseline Classification Report (Churn Prediction)

Date: 26 April 2026

Dataset Used: Churn_Modelling.csv

Dataset Selection:

The selected dataset is well-suited for a classification problem as it contains a binary target variable (Exited) representing whether a customer has churned or not. It includes both numerical and categorical features such as age, balance, geography, and activity status, which are relevant for predicting customer behavior. The dataset is structured, clean, and widely used for benchmarking classification models, making it appropriate for implementing Logistic Regression as a baseline model.

Objective:

The objective of this task is to develop a baseline classification model using Logistic Regression to predict customer churn. Additionally, the goal is to interpret feature importance through model coefficients (log-odds) and evaluate model performance using metrics such as Confusion Matrix, Precision, Recall, and F1-score.

Task Performed:

1. Loaded the dataset and examined its structure and data types.
2. Removed irrelevant columns such as RowNumber, CustomerId, and Surname.
3. Encoded categorical variables:

    - Converted Gender using Label Encoding
    - Applied One-Hot Encoding on Geography

4. Defined input features (X) and target variable (y).
5. Split the dataset into training and testing sets (80:20 ratio).
6. Applied **Standard Scaling** to normalize feature values.
7. Trained a **Logistic Regression model** as the baseline classifier.
8. Extracted feature coefficients to interpret their influence on churn prediction.

9. Generated predictions on the test dataset.
10. Evaluated model performance using:

    - Confusion Matrix
    - Precision
    - Recall
    - F1-score

Conclusion:

The Logistic Regression model successfully established a baseline for the churn prediction task. It provided interpretable results through feature coefficients, highlighting the impact of key variables such as age, balance, and customer activity on churn behavior. While the model offers simplicity and interpretability, its performance can be further improved by using advanced machine learning algorithms and addressing potential class imbalance.